UNIVERSALLY INJECTIVE STRUCTURE MORPHISM IN THE VANISHING SITUATION, BY SEPARATING THE CLOSED POINTS OF TWO RATIONAL POINTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Universal injectivity may be checked after an arbitrary field valued base change. Morphisms of Spaces, label spaces-morphisms-lemma-universally-injective-local (invoked at proof line 3106). It reduces the claim to showing that for every field valued point the base change has at most one point.
- The vanishing situation is stable under base change by an algebra, and separation of disjoint closed subsets. Same chapter spaces-cohomology.tex, labels lemma-vanishing-base-change (invoked at proof line 3107) and lemma-vanishing-separate-closed (invoked at proof line 3127). The first lets the argument assume that the base ring is a field; the second produces a global function separating two distinct closed points, which forces the two k-valued points to coincide.
- Points of an algebraic space over a large algebraically closed extension, and rational points as closed points. Morphisms of Spaces, labels spaces-morphisms-lemma-large-enough (invoked at proof line 3117) and Decent Spaces, label decent-spaces-lemma-algebraic-residue-field-extension-closed-point (invoked at proof line 3124). The first makes $X(K) \to |X|$ surjective for a large enough algebraically closed extension, reducing the claim to singleton sets of rational points; the second makes rational points closed, so that the separating function applies to them.

Source reference. Original lemma 07U5: sources/stacks-chow/tags.txt, source lines 9960–9960. Navigation pointer for \ref{decent-spaces-lemma-algebraic-residue-field-extension-closed-point} used inside the counted span (line 3124); the argument that consumes it is retained in the span.

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Source reference. Original lemma 07V0: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3041–3041. Navigation pointer for \ref{lemma-vanishing-base-change} used inside the counted span (line 3107); the argument that consumes it is retained in the span.

Source reference. Original lemma 07V1: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3072–3072. Navigation pointer for \ref{lemma-vanishing-separate-closed} used inside the counted span (line 3127); the argument that consumes it is retained in the span.

Source reference. Original reference 07UY: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2991–2991. Navigation pointer for \ref{situation-vanishing} used inside the counted span (line 3098); the argument that consumes it is retained in the span.

Source reference. Original lemma 0488: sources/stacks-chow/tags.txt, source lines 5535–5535. Navigation pointer for \ref{spaces-morphisms-lemma-large-enough} used inside the counted span (line 3118); the argument that consumes it is retained in the span.

Source reference. Original lemma 03MX: sources/stacks-chow/tags.txt, source lines 4765–4765. Navigation pointer for \ref{spaces-morphisms-lemma-universally-injective-local} used inside the counted span (line 3106); the argument that consumes it is retained in the span.

Source reference. Original section 0485: sources/stacks-chow/tags.txt, source lines 5532–5532. Navigation pointer for \ref{spaces-morphisms-section-points-fields} used inside the counted span (line 3114); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 07UY the morphism $p : X \to \mathrm{Spec}(A)$ is universally injective.*

Proof. Let $A \to k$ be a ring homomorphism where k is a field. It suffices to show that $\mathrm{Spec}(k) \times_{\mathrm{Spec}(A)} X$ has at most one point (see Morphisms of Spaces, Lemma 03MX). Using Lemma 07V0 we may assume that A is a field and we have to show that $|X|$ has at most one point.

Let's think of X as an algebraic space over $\mathrm{Spec}(k)$ and let's use the notation $X(K)$ to denote K-valued points of X for any extension K/k, see Morphisms of Spaces, Section 0485. If K/k is an algebraically closed field extension of large transcendence degree, then we see that $X(K) \to |X|$ is surjective, see Morphisms of Spaces, Lemma 0488. Hence, after replacing k by K, we see that it suffices to prove that $X(k)$ is a singleton (in the case $A = k$).

Let $x, x' \in X(k)$. By Decent Spaces, Lemma 07U5 we see that x and x' are closed points of $|X|$. Hence x and x' map to distinct points of $\mathrm{Spec}(k)$ if $x \neq x'$ by Lemma 07V1. We conclude that $x = x'$ as desired. □